Designing and Simulation of Circuits with Python and Generative AI

With the help of generative AI, Python libraries like SchemDraw, ngpsice and pandapower are simplifying the design and simulation of electrical and electronic circuits for research.

Generative AI is the buzzword today, with most companies trying to adopt it to automate their work, bring in a higher degree of accuracy in operations, and increase their productivity.

According to GrandViewResearch, the global market size of generative AI was US$ 16.8 billion in 2024 and is projected to reach US$ 109.4 billion by 2030. A report from MarketsAndMarkets.com forecasts the generative AI market to exceed US$ 890 billion by 2032.

The key application areas of generative AI are:
- Text summarisation
- Circuit design and simulation in engineering
- Image generation and image style transfer
- Music composition and voice cloning
- Video creation
- Code generation
- Chat automation
- Marketing copywriting
- Content rewriting
- Forecasting and analytics of financial data
- Product design
- Data augmentation
- Game asset creation
- Financial forecasting
- Medical report analytics and predictions
- Language translation
- Presentations and slide shows

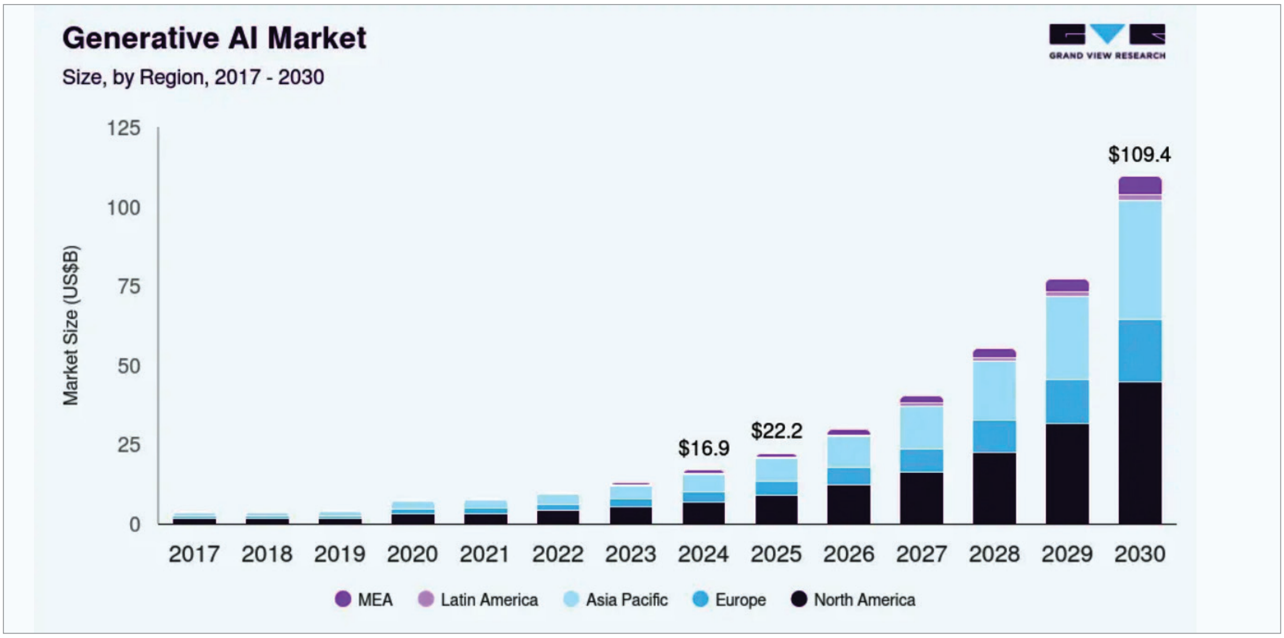

Figure 1: Market size of generative AI (Source: GrandViewResearch)